OIL & GAS TECHNICAL ARCHITECTURE BRIEF — UPSTREAM • MIDSTREAM • DOWNSTREAM

Rouen for the Oil & Gas Industry

Connecting remote multi-well pads, pipeline stations, and analyzer shelters to central production & gas control rooms via Outbound-Only WebSocket Mesh and Zero-Compile Adaptive Process RTU Bridges.

API RP 1164 & ISA/IEC 62443 Ready | Starlink / VSAT / Cellular CGNAT | Enron Modbus • ROC • HART • WITS0 | Ed25519 Mutual Auth | Class I Div 2 Edge Ready

01 The Oilfield & Pipeline OT/IT Challenge

Upstream wellpads, midstream gathering pipelines and compressor stations, and downstream LNG/refinery facilities face two critical barriers when modernizing production surveillance, artificial lift optimization, and remote custody transfer:

1. Extreme Remote Connectivity & API RP 1164 Isolation

- **Unmanned CGNAT & Satellite Sites:** Multi-well pads, pipeline block valves, and offshore platforms connect via Starlink, VSAT, BGAN, or cellular modems with no public IP address and strict bandwidth caps.
- **Purdue Model & API RP 1164 Firewalls:** Pipeline and facility cybersecurity mandates prohibit inbound connections from corporate IT or cloud networks into Level 1/2 RTU and PLC zones.
- **VPN Sprawl Across Lease Operators:** Maintaining hundreds of brittle IPsec tunnels across dynamic cellular/satellite links drains field OT teams.

2. Oilfield Protocol & Legacy RTU Fragmentation

- **Specialized Oilfield Protocols:** Field skids speak Enron Modbus, Emerson ROC, Bristol BSAP, ABB Totalflow, HART-IP, OPC-UA, and WITS0/WITSML drilling streams.
- **Rigid SCADA & DCS Change Cycles:** Adding an ESP variable speed drive, H2S detector, or gas chromatograph often requires expensive SCADA tag packs and vendor integrator weeks.
- **Harsh Edge Hardware Limits:** DIN-rail gateways inside NEMA 4X wellpad enclosures cannot run bloated JVM or container clusters.

02 Pillar 1: Outbound-Only WebSocket Mesh (`rouen_mesh_host`)

Implemented in `rouen::mesh::rouen_mesh_host`, every field gateway connects to the Rouen Mesh via a single outbound TLS WebSocket (`wss://` on port 443). Unmanned wellpads and compressor stations never open listening ports. Identity is cryptographically verified via mutual Ed25519 challenge-response signatures, multiplexing live telemetry cards (`HTTP_REQUEST` / `HTTP_RESPONSE`), well/pipeline alarm registries (`REGISTRY_SET`), and auditable diagnostic streams (`ROUTE_OPEN`).

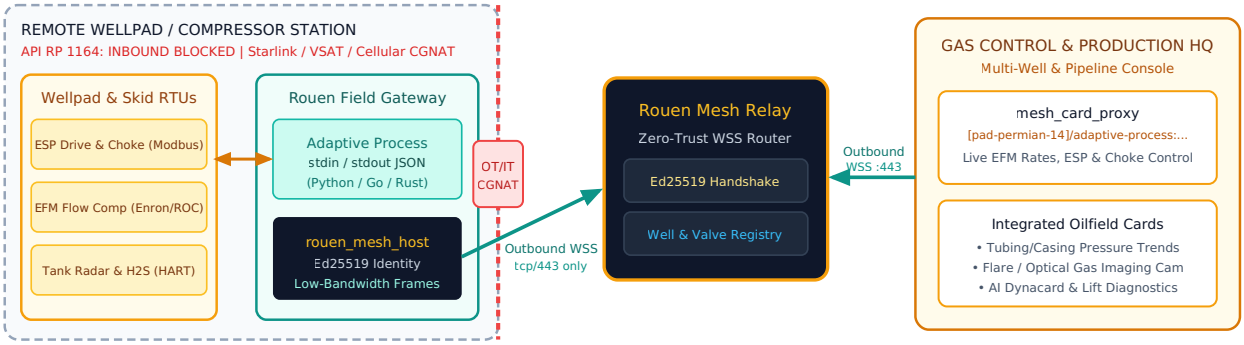


Figure 1 — Zero-inbound-port Oil & Gas topology: Wellpad gateways behind Starlink/cellular CGNAT initiate outbound-only WSS connections. Raw Enron Modbus, ROC, and HART traffic never leaves the wellpad.

03 Pillar 2: Universal RTU & Sensor Bridge (`adaptive_process_card`)

Implemented in `rouen::cards::adaptive_process_card` on top of `rouen::helpers::piped_process`, the `adaptive-process:<command>` URI scheme turns any script or executable in any language into a live, interactive wellpad or pipeline card without compiling against Rouen:

- `stdout` Stream (Live RTU/EFM Telemetry): Each compact JSON line printed by the field script renders immediately as an Adaptive Card UI, with automatic cadence polling via `"refreshIntervalMs"`.
- `stdin` Stream (Choke, ESP & Valve Commands): Operator actions (`Action.Execute`, such as updating an automated choke opening, adjusting ESP Hz, or triggering an SSV shut-in) serialize form inputs as a single JSON line to the script's `stdin`.
- Process Isolation & Live `stderr` Diagnostics: Noisy RS-485 serial loops or hung vendor drivers are isolated in the child OS process, with diagnostic logs captured in a collapsible Process stderr drawer.

04 The "Killer Combo": Mesh Card Proxy + Adaptive Process on the Wellpad

DIRECT EDGE EXECUTION WITH CENTRAL PRODUCTION CONTROL

By composing `mesh_card_proxy` with `adaptive_process_card`, a production engineer or gas controller opens a live interactive card backed by a protocol script running directly on the wellpad gateway:

```
[pad-permian-14]/adaptive-process:python3 /opt/oilfield/esp_wellhead_bridge.py --well 3H
```

Why it wins in Oil & Gas: High-frequency RS-485/Modbus polling stays local on the pad (saving satellite/cellular data), while engineers get instant, cryptographically authenticated choke and lift control from any paired Rouen console.

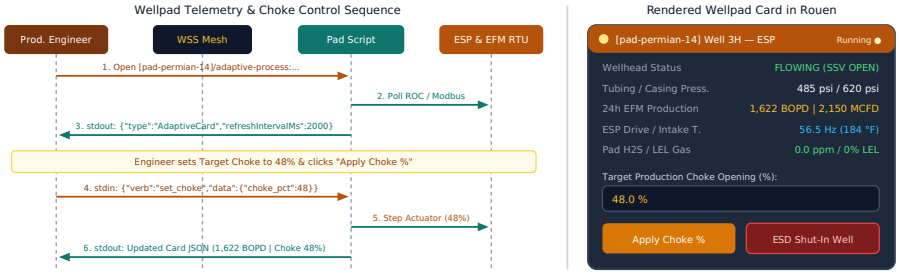


Figure 2 — Left: Bidirectional stdio-over-Mesh protocol flow. Right: The resulting live Well 3H ESP & Choke Card rendered in Rouen.

05 Reference Implementation: Python Wellhead ESP & Choke Bridge

A complete wellhead RTU/Modbus bridge requires ~35 lines of Python. Any language capable of reading `stdin` and printing newline-delimited JSON to `stdout` works identically:

```python
#!/usr/bin/env python3  —  Launch remotely: [pad-permian-14]/adaptive-process:python3 /opt/oilfield/esp_wellhead_bridge.py
import json, sys, time
w = {"well": "Permian Pad 14 — Well 3H (ESP + Choke)", "tbg": 485, "csg": 620, "bopd": 1420, "mcfd": 2150, "hz": 56.5, "choke": 42.0, "ssv": True}

def emit_card(note="Nominal"):
    status = "FLOWING (SSV OPEN)" if w["ssv"] else "SHUT-IN (SSV CLOSED)"
    card = {"type": "AdaptiveCard", "version": "1.5", "refreshIntervalMs": 2000, "body": [
        {"type": "TextBlock", "text": w["well"], "size": "large", "weight": "bolder"},
        {"type": "FactSet", "facts": [
            {"title": "Wellhead Status",    "value": status},
            {"title": "Tubing / Casing",    "value": f"{w['tbg']} psi / {w['csg']} psi"},
            {"title": "24h EFM Rates",      "value": f"{w['bopd']:.0f} BOPD | {w['mcfd']} MCFD"},
            {"title": "ESP Drive / Choke",  "value": f"{w['hz']:.1f} Hz | Choke: {w['choke']:.1f}% ({note})"}]},
        {"type": "Input.Number", "id": "choke", "title": "Target Choke Opening (%)", "value": w["choke"]}
    ], "actions": [
        {"type": "Action.Execute", "title": "Apply Choke %", "verb": "set_choke"},
        {"type": "Action.Execute", "title": "ESD Shut-In Well", "verb": "toggle_ssv"}]}
    print(json.dumps(card), flush=True)

emit_card("Initial RTU scan")
for raw in sys.stdin:
    msg = json.loads(raw.strip() or "{}"); verb, data = msg.get("verb", ""), msg.get("data", msg)
    if verb == "set_choke": w["choke"] = max(0.0, min(100.0, float(data.get("choke", w["choke"]))))
    elif verb == "toggle_ssv": w["ssv"] = not w["ssv"]
    w["bopd"] = (w["choke"] * 33.8) if w["ssv"] else 0.0; emit_card(verb or "Live RTU poll")
```

06 High-Value Oil & Gas Use Cases

OIL & GAS SEGMENT	OPERATIONAL PAIN POINT	ROUEN MESH + ADAPTIVE PROCESS ADVANTAGE
Upstream: Artificial Lift & Multi-Well Pads	Hundreds of ESP, Gas Lift, and Rod Pump wells behind cellular CGNAT or Starlink satellite modems.	Outbound-only `wss://` links every wellpad to production engineers with zero VPN tunnels or inbound ports.
Midstream: Pipelines & Compressor Stations	Metered VSAT links and strict API RP 1164 isolation across remote block valves, scraper traps, and EFM skids.	Ultra-lightweight single-line JSON updates and authenticated valve/setpoint control over persistent binary frames.
Drilling & Completions (Rig & Frac Spreads)	Aggregating WITS0/WITSML mud logging, frac iron pressures, and live cameras across mobile rig sites.	Unifies Rouen's native Number Series and Camera cards alongside Adaptive Process WITS0/Modbus bridges.
Downstream: Refinery & LNG Analyzer Shelters	Integrating gas chromatographs, H2S/LEL sensors, and vibration rigs without modifying core DCS safety logic.	Isolated child-process scripts provide side-channel engineering diagnostics with zero impact on DCS scan times.